

Key Outputs

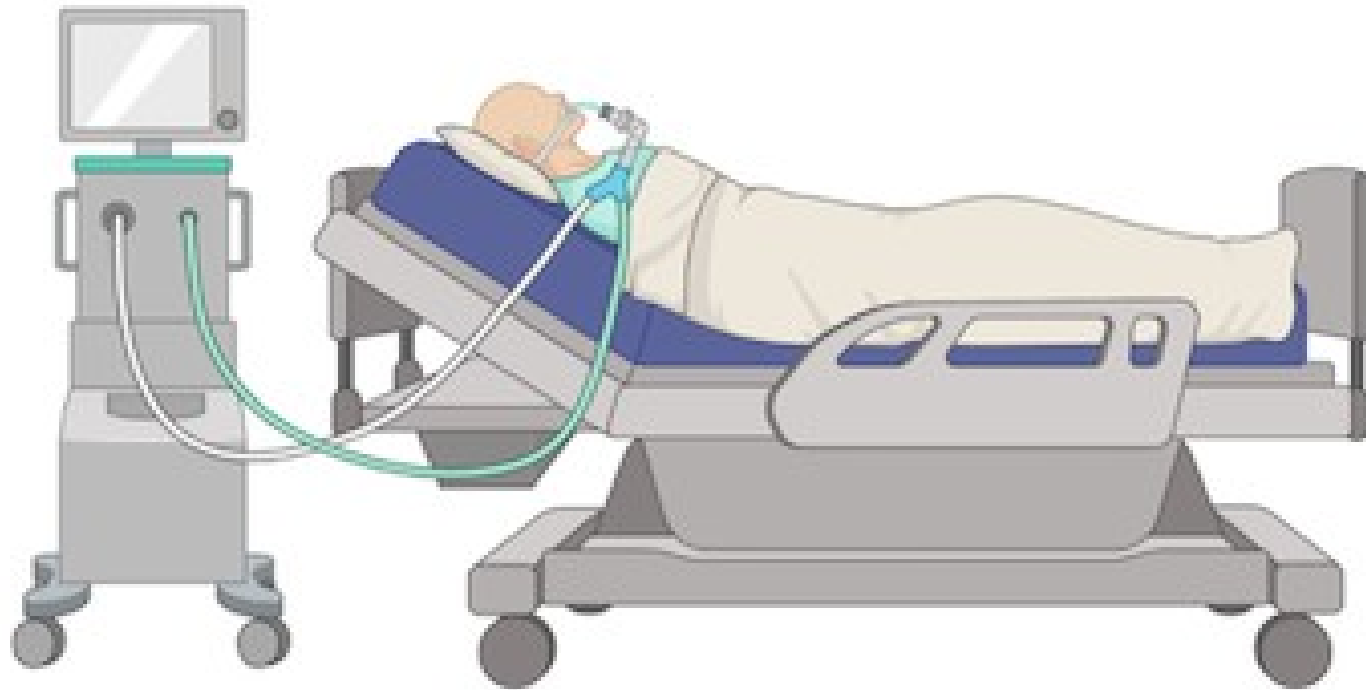
Definitions

Peak P Alarm

Summary

Cases

Assessing Pulmonary Mechanics in a Ventilated Patient



Key Outputs

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Peak P Alarm

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Key Outputs to Monitor Pulmonary Mechanics

Pressure-Targeted Modes

Key Parameters?

Tidal volume (V_T)

Minute ventilation (V_E)

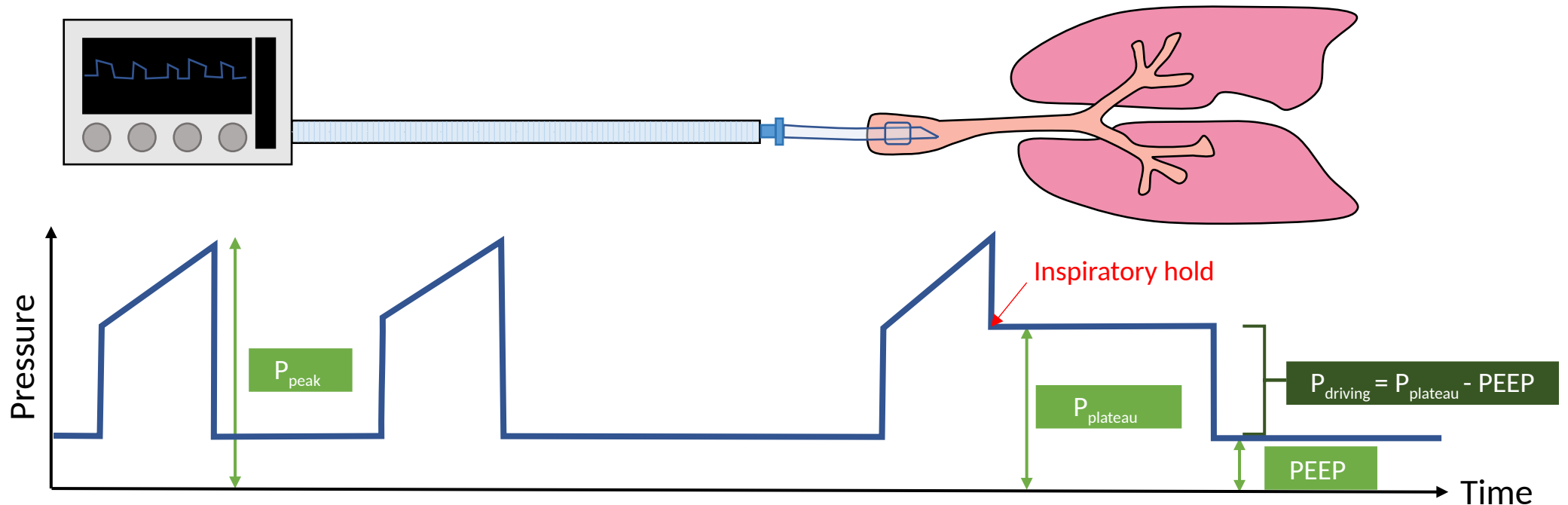
Volume-Targeted Modes

Key Parameters?

Peak pressure (P_{peak})

Plateau pressure ($P_{plateau}$)

Driving pressure ($P_{driving}$)



Key Outputs

Definitions

P_{peak}

P_{plateau}

P_{driving}

Peak P Alarm

Summary

Cases

Definitions

RESISTANCE

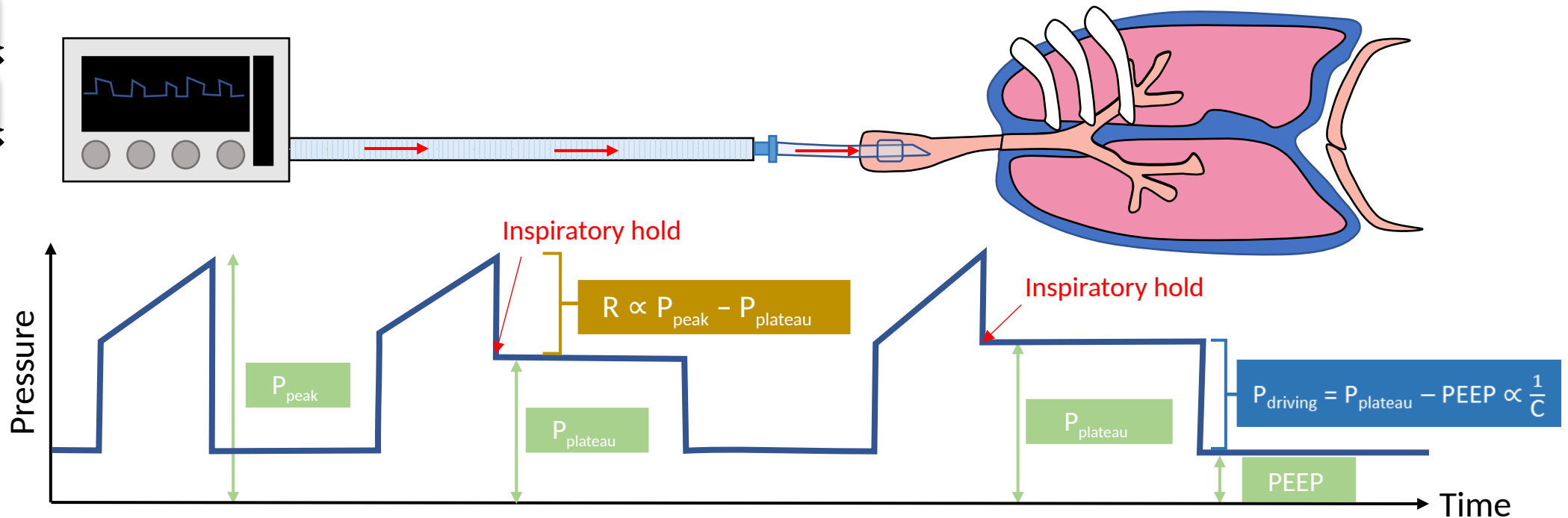
- Opposition to airflow through the system (tubing, airways)
- Typically, <10.

COMPLIANCE

- Stiffness of respiratory system (lungs, pleural, chest wall, abdomen)

$$C = \frac{V_T}{P_{\text{plateau}} - \text{PEEP}} = \frac{V_T}{P_{\text{driving}}}$$

- Normally, pressure needed to achieve 4-6 cc/kg is <10



Key Outputs

Definitions

P_{peak}

P_{plateau}

P_{driving}

Peak P Alarm

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Cases

What Influences Peak Pressures?

1

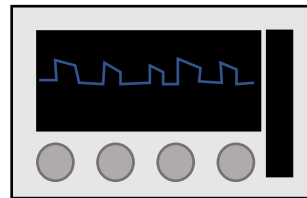
RESISTANCE

- Opposition to airflow through the system (tubing, airways)
- Typically, <10.
- $\uparrow$ resistance $\rightarrow \uparrow$ peak pressure

2

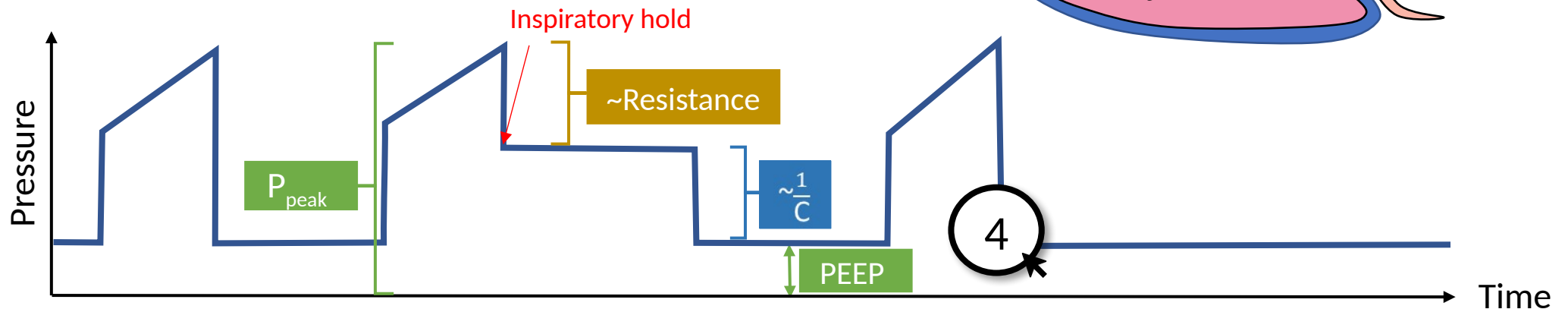
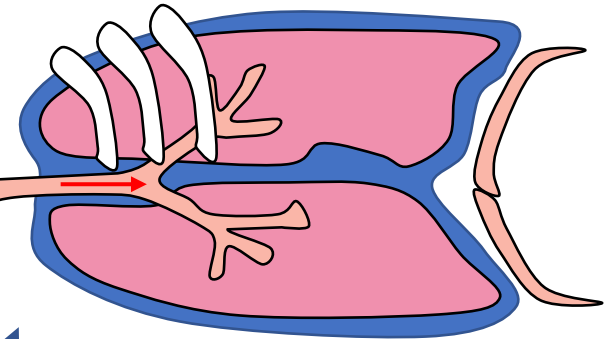
COMPLIANCE

- Stiffness of respiratory system (lungs, pleural, chest wall, abdomen)
- Normally, pressure needed to achieve 4-6 cc/kg is <10



3

Inspiratory flow rate/ pattern



Key Outputs

Definitions

P_{peak}

P_{plateau}

P_{driving}

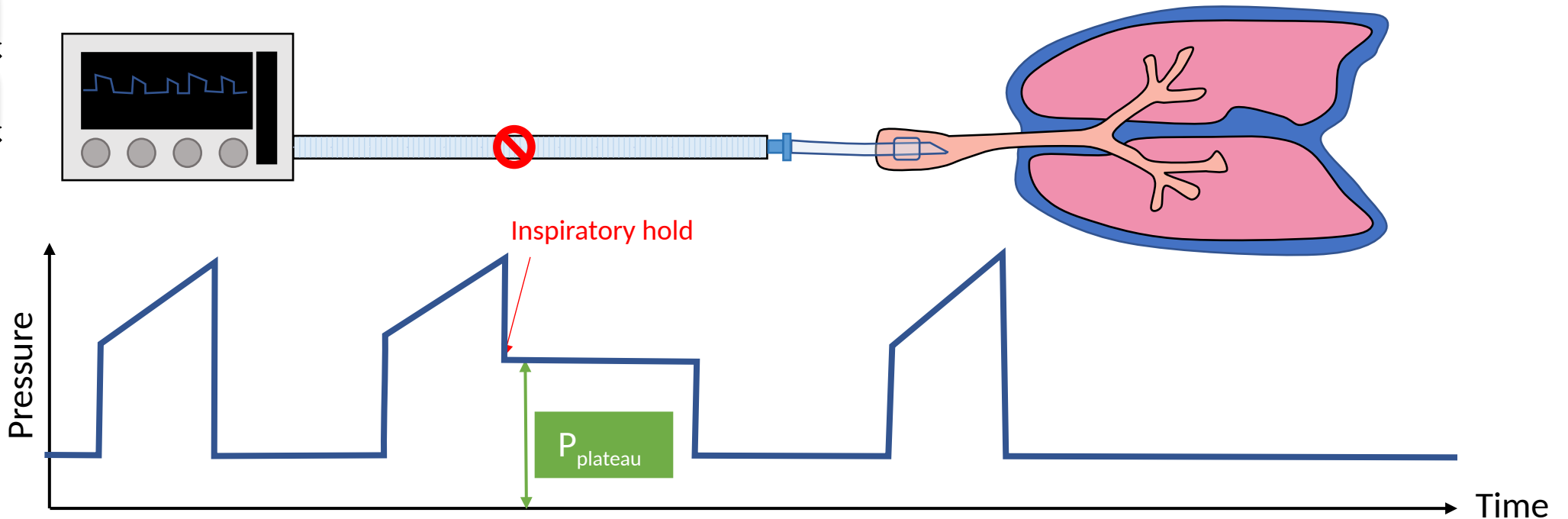
Peak P Alarm

Summary

Cases

What Influences Plateau Pressure?

Because plateau pressure is measured during an inspiratory hold, there is NO AIR FLOW and thus no resistance in the system.



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Definitions

P_{peak}

P_{plateau}

P_{driving}

Peak P Alarm

Summary

Cases

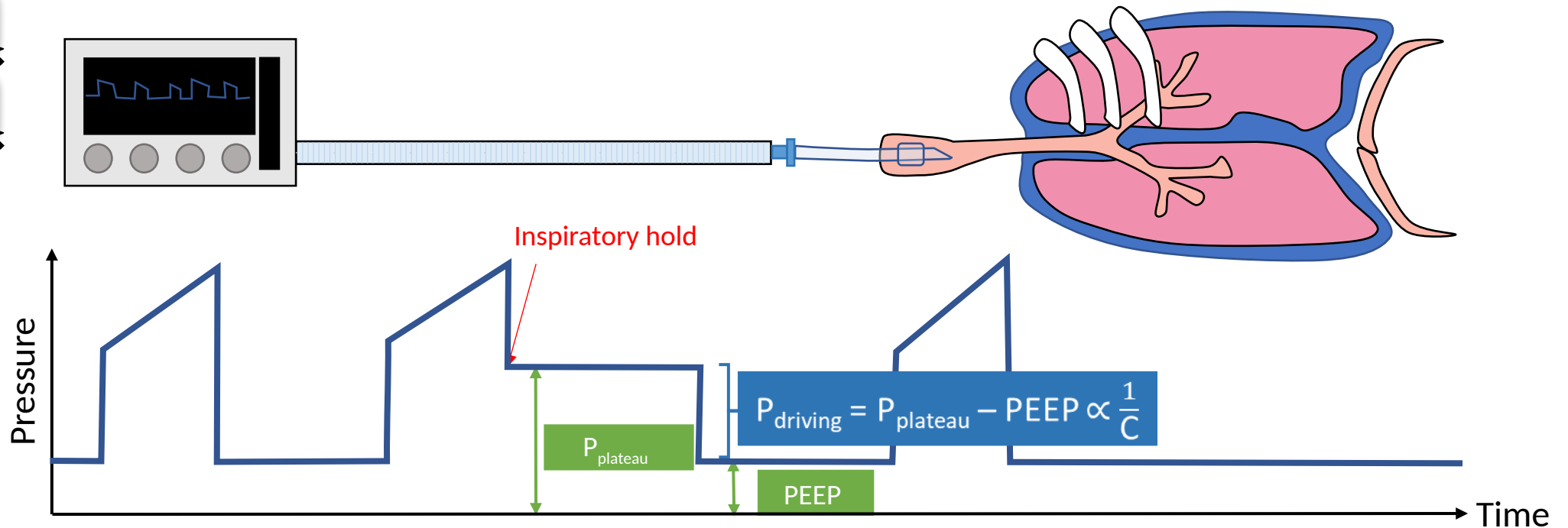
What Influences Driving Pressure?

COMPLIANCE

- Stiffness of respiratory system (lungs, pleural, chest wall, abdomen)

$$C = \frac{V_T}{P_{\text{plateau}} - PEEP} = \frac{V_T}{P_{\text{driving}}}$$

- Normally, pressure needed to achieve 4-6 cc/kg is <10



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Peak P Alarm

Step 1

Step 2

Step 3

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Responding to Peak Pressure Alarm

STEP 1: ESTABLISH URGENCY

Check vitals

P_{CO_2} (on ABG)

V_T and V_E (on vent)

$\downarrow O_2, V_T, V_E$

or

$\uparrow P_{\text{CO}_2}$

Get HELP!



$O_2 > 88\%$

$V_T > 4\text{cc/kg}$

$\downarrow \downarrow V_E$

Proceed



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Peak P Alarm

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Responding to Peak Pressure Alarm

STEP 1: ESTABLISH URGENCY

Proceed if:
Hemodynamically stable

$O_2 > 88\%$

$V_T > 4\text{cc/kg}$

$\downarrow \downarrow V_E$

STEP 2: ASSESS RESISTANCE

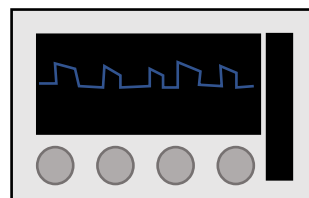
Peak Pressure $\uparrow$

Plateau Pressure $\leftrightarrow$

$P_{\text{peak}} - P_{\text{plateau}} > 10$ suggests
resistance problem

Differential?

Action?

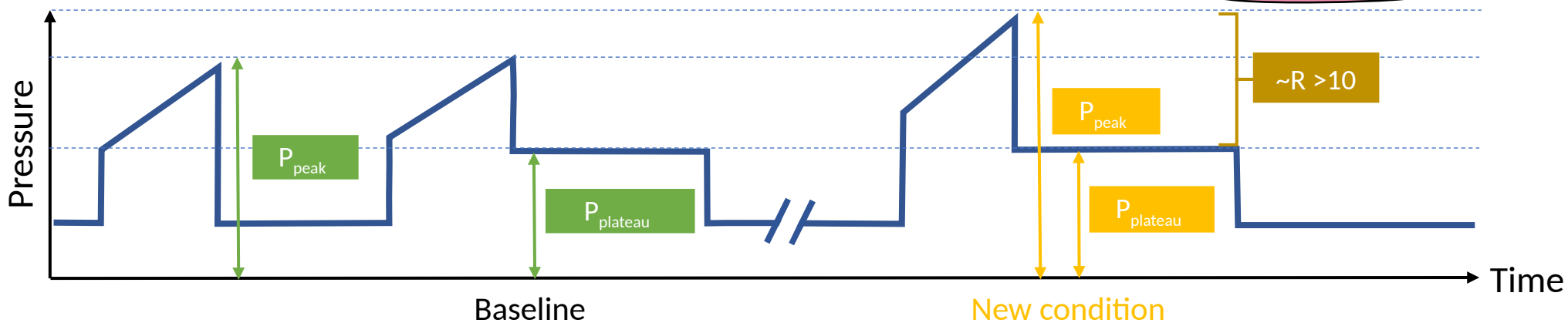


Kinked tubing or ETT

Biting ETT

Mucus plugging

Bronchospasm



Key Outputs

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Responding to Peak Pressure Alarm

STEP 1: ESTABLISH URGENCY

Proceed if:
Hemodynamically stable

$O_2 > 88\%$

$V_T > 4\text{cc/kg}$

$\downarrow \downarrow V_E$

STEP 2: ASSESS RESISTANCE

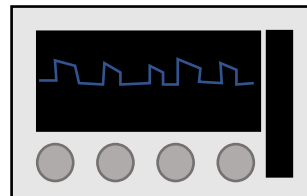
Peak Pressure $\uparrow$

Plateau Pressure $\leftrightarrow$

$P_{\text{peak}} - P_{\text{plateau}} > 10$ suggests
resistance problem

Differential?

Action?



Kinked tubing or ETT

Check tubing

$\uparrow$ analgesia/ sedation

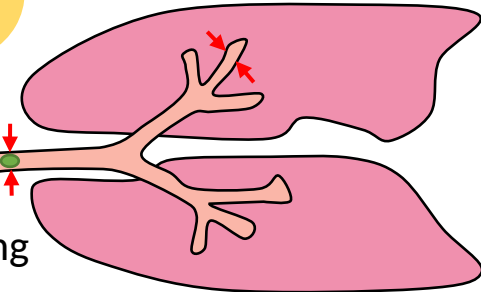
Biting ETT

Mucus plugging

Suction

Bronchospasm

Bronchodilators



Key Outputs

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Responding to Peak Pressure Alarm

STEP 1: ESTABLISH URGENCY

Proceed if:
Hemodynamically stable

$O_2 > 88\%$

$V_T > 4\text{cc/kg}$

$\downarrow \downarrow V_E$

STEP 2: ASSESS RESISTANCE

Peak Pressure $\uparrow$

Plateau Pressure $\leftrightarrow$

$P_{\text{peak}} - P_{\text{plateau}} > 10$ suggests
resistance problem

Differential?

Bronchospasm

Kinked tubing

Mucus plug

STEP 3: ASSESS COMPLIANCE

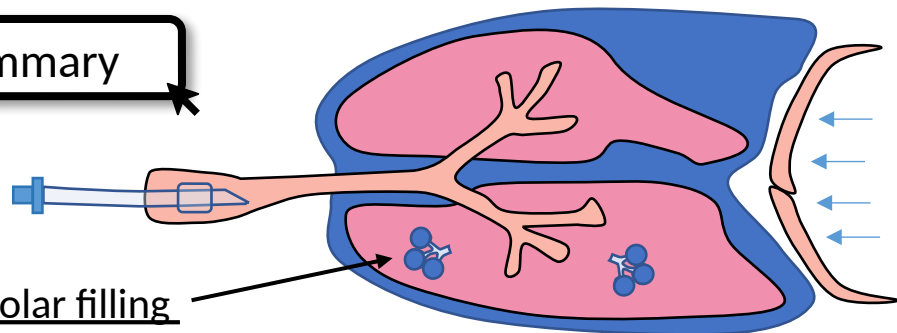
Peak Pressure $\uparrow$

Plateau Pressure $\uparrow$

$P_{\text{driving}} > 10$ suggests compliance
problem

Differential?

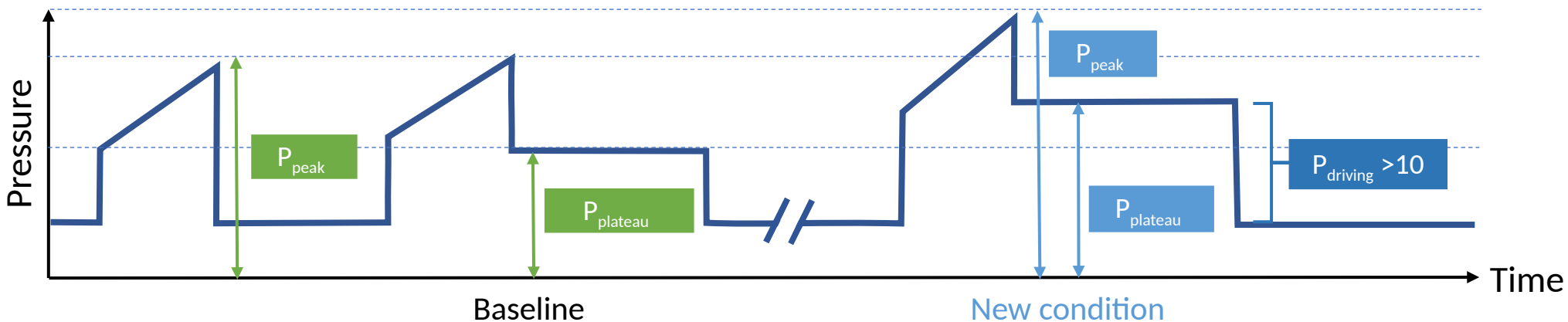
Action?



- ARDS
- Pulmonary edema
- Pneumonia
- DAH

Alveolar collapse

- PTX
- Abdominal distention
- Atelectasis/ lobar collapse
- Pleural effusion



Key Outputs

Definitions

Peak P Alarm

Step 1

Step 2

Step 3

Summary

Responding to Peak Pressure Alarm

STEP 1: ESTABLISH URGENCY

Proceed if:
Hemodynamically stable
 $O_2 > 88\%$
 $V_T > 4\text{cc/kg}$
 $\downarrow \downarrow V_E$

STEP 2: ASSESS RESISTANCE

Peak Pressure $\uparrow$
Plateau Pressure $\leftrightarrow$
 $P_{\text{peak}} - P_{\text{plateau}} > 10$ suggests
resistance problem

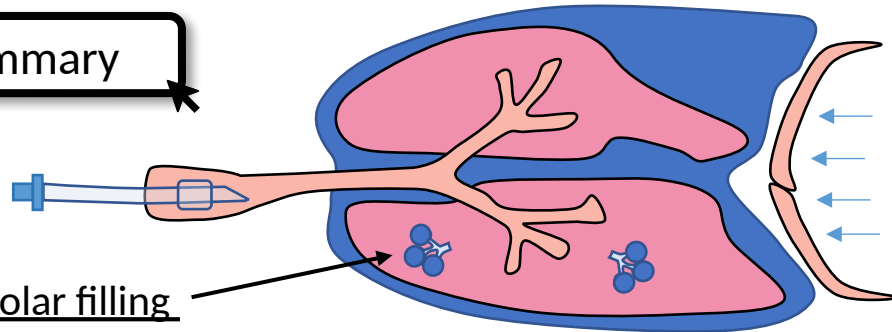
Differential?
Bronchospasm
Kinked tubing
Mucus plug

STEP 3: ASSESS COMPLIANCE

Peak Pressure $\uparrow$
Plateau Pressure $\uparrow$
 $P_{\text{driving}} > 10$ suggests compliance
problem

Differential?

Action?



- ARDS
- Pulmonary edema
- Pneumonia
- DAH

- Evaluate for underlying pathology
 - Exam
 - CXR
 - Abdominal XR
- Treat underlying pathology

Alveolar collapse

- PTX
- Abdominal distention
- Atelectasis/ lobar collapse
- Pleural effusion

Key Outputs

Definitions

Peak P Alarm

Summary

Cases

Responding to Peak Pressure Alarm

STEP 1: ESTABLISH URGENCY

Proceed if:
Hemodynamically stable

$O_2 > 88\%$

$V_T > 4\text{cc/kg}$

$V_E > 3\text{L/min}$

STEP 2: ASSESS RESISTANCE

Peak Pressure $\uparrow$

Plateau Pressure $\leftrightarrow$

$P_{\text{peak}} - P_{\text{plateau}} > 10$ suggests
resistance problem

Differential?

Bronchospasm

Kinked tubing

Mucus plug

STEP 3: ASSESS COMPLIANCE

Peak Pressure $\uparrow$

Plateau Pressure $\uparrow$

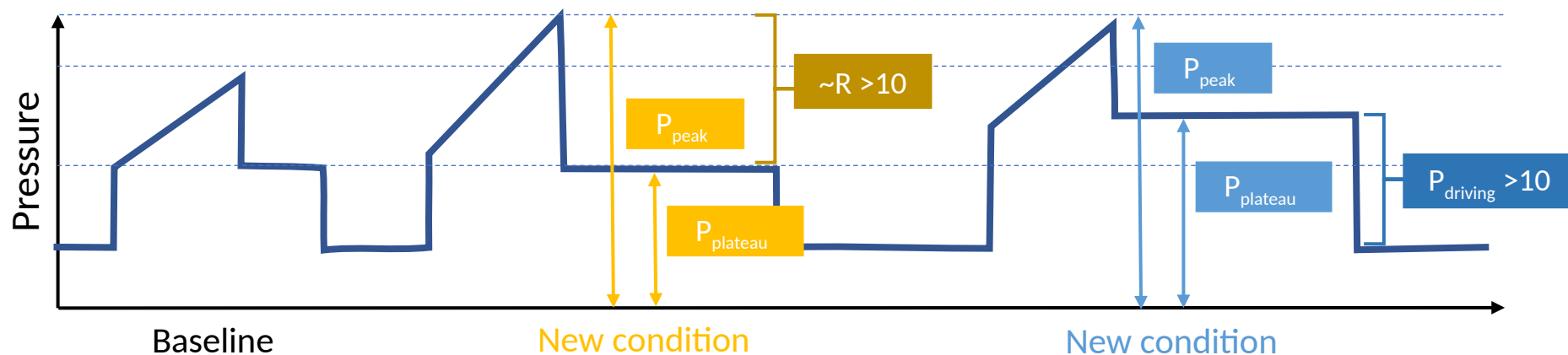
$P_{\text{driving}} > 10$ suggests compliance
problem

Differential?

Intrapulmonary process

Pleural effusion

Abdominal distention



Key Outputs

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Peak P Alarm

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Cases

1

2

Case 1

65 yo is intubated in the ICU for COVID pneumonia. You are called to assess gradually rising P_{peak} , which increased from ~ 35 to consistently 40. He is on assist control, volume control mode of ventilation. Settings are: 70% FiO_2 , PEEP 12, RR 20, V_t 5 cc/kg (450 cc). The patient is currently deeply sedated and paralyzed.

Vitals: T 37C, BP 110/57, HR 93, RR 24, SaO₂ 97%. V_t is 450 cc.

What is your first step in evaluation?

Establish urgency. Vitals and oxygenation are stable. V_E (RR x V_t) is acceptable. This patient is stable.

What is your next step in evaluation?

Assess for a resistance and/or compliance problem **by obtaining a plateau pressure**. This can be done with an inspiratory hold.

...Plateau pressure is 38 cmH₂O.

Is there a resistance problem?

No. Resistance is proportional to Peak – Plateau. A difference >10 suggests high resistance. Here, 40 – 38 = 2, which is low.

Is there a compliance problem?

Yes. Compliance is inversely proportional to the driving pressure (Plateau – PEEP). The driving pressure (38 – 12) is 26, which is elevated. This suggests low compliance.

What is your differential for the low compliance?

- Alveolar process - pulmonary edema, worsening pneumonia, atelectasis
 - Restricted expansion - pleural effusions
- Given slow rise in peak pressure, a pneumothorax or lobar collapse are less likely.

What are your next steps in evaluation?

Get additional history. Perform a targeted physical exam. Obtain a **CXR** or bedside US and **ABG** to evaluate for new intrapulmonary processes. CBC, BMP, BNP.

...Exam reveals soft abdomen, diffuse crackles. CXR shows worsening bilateral interstitial and alveolar infiltrates. CBC shows worsening leukocytosis. pH 7.34/45/82/24 on 70% FiO_2 .

What are the most likely diagnosis?

Worsening parenchymal lung disease, which could include from ARDS or superimposed bacterial infection.

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1

2

Case 2

25 yo with asthma presents with acute hypercarbic and hypoxemic respiratory failure from an asthma exacerbation. They are intubated and placed on volume-targeted mode of ventilation with FiO₂ of 70%, PEEP 5, V_t 350 cc (6cc/kg), RR 24. One hour after intubation, they become progressively hypotensive and hypoxemic. BPs are 100s/60s, RR 30, SaO₂ 92%, HR 120s. The ventilator is alarming for high peak pressures of 38 cm H₂O. V_t delivered is 300 cc.

What is the urgency of the situation?

Vitals and oxygenation are stable. Proceed with assessment. Obtain ABG to assess pCO₂ and pH if not done recently.

What is your next step in management?

Inspiratory hold to measure the **plateau P**. This helps differentiate a resistance vs. compliance issue.

...The plateau P is 20. ABG shows 7.25/60/95/24.

Is there a resistance or compliance problem?

Resistance. Peak - Plateau is = 15. There is NO compliance issue. Plateau is <30 and driving P (Plateau - PEEP) is <10.

What is the most likely diagnosis?
How would you manage this?

Bronchospasm and airway inflammation.
Escalate bronchodilator ± steroid therapy. Consider IV magnesium if no improvement with aggressive bronchodilators.

...After escalating therapy and ↑ RR to address the acidosis, the Peak P is still 35. Plateau P ↑ to 30. RR is 35, PEEP 12. BPs 80s/60s.

Is there a resistance or compliance problem?

The patient's BPs are trending down. **Call for help!**
There is a **compliance** issue. Plateau P - PEEP >10.
There is NO resistance issue. Peak - Plateau <10.

What is the most likely diagnosis?

AutoPEEP or pneumothorax.

What is your next step in evaluation/management?

Slow RR to ↑ expiration time (as long as pH allows). Listen for bilateral breath sounds. **CXR** or **POCUS** to evaluate for PTX.

...The patient goes into cardiac arrest. POCUS rules out PTX.

What is your next step in management?

Call a Code Blue. Disconnect from ventilator and allow for long exhalation. Start ACLS and use manual bagging (RR 8-10).

Pulmonary Mechanics — 2023–25 Context Update

Core mechanics teaching (resistance, compliance, peak/plateau/driving pressure, the high-peak-pressure algorithm) is timeless physiology — verified current June 2026. The items below add context, not corrections.

Driving pressure ($\Delta P = P_{\text{plateau}} - PEEP$) is the mechanic most tightly linked to ARDS survival. Keep $\Delta P \leq 14\text{--}15$ cm H₂O — risk rises stepwise above this even when $V_t \leq 6$ mL/kg and $P_{\text{plat}} \leq 30$ (Amato, NEJM 2015; LUNG SAFE).

Lung-protective targets unchanged: V_t 4–6 mL/kg ideal body weight, $P_{\text{plateau}} < 30$ cm H₂O, $PEEP > 5$ (ARDSNet NEJM 2000; ATS/ESICM/SCCM 2017; ESICM ARDS 2023).

ESICM ARDS 2023 / Global (new) ARDS definition 2024: $SpO_2/FiO_2 \leq 315$ (when $SpO_2 \leq 97\%$) may define ARDS, lung ultrasound is an accepted imaging modality, and HFNO is included as a recognized support category — broadens who you will be troubleshooting on the vent.

Bedside rule of thumb retained: normal resistance $P_{\text{peak}} - P_{\text{plateau}} < 10$; normal compliance driving pressure < 10 cm H₂O for a 4–6 mL/kg breath.

Take-Home Points

1. Peak and plateau are the key outputs in volume-targeted modes. Peak pressure reflects resistance + compliance + PEEP + inspiratory flow; plateau (measured on an inspiratory hold) should be <30 cm H₂O.
2. For a high-peak-pressure alarm: (1) establish urgency at the bedside, then (2) get a plateau pressure and decide resistance vs compliance vs both.
3. Resistance $\propto$ P_{peak} – P_{plateau}. >10 cm H₂O = resistance problem → kinked/bitten/blocked tube, mucus plug, bronchospasm. Fix: inspect tubing, suction, bite block/sedation, bronchodilators.
4. Compliance $\propto$ 1/driving pressure (P_{plateau} – PEEP). Driving pressure >10 (or P_{plateau} >30) = compliance problem → alveolar filling (edema, ARDS, pneumonia, DAH) or collapse (atelectasis, PTX, effusion, abdominal distention). Keep $\Delta P \leq 14-15$.
5. They can co-exist — and a falling BP, SpO₂, or VE turns troubleshooting into an emergency: call for help and bump FiO₂ to 100% first.

Attribution & Optimization Notes

Original lesson: “Ventilators Part 2 — Assessing Pulmonary Mechanics in a Ventilated Patient,” TeachIM.org (updated Apr 2023).

Authors: Kaitlyn McLeod MD, David Scudder MD; Section Editor Trevor Steinbach MD; Executive Editor Yilin Zhang MD.

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Correction made (this edit): Case 1 driving-pressure arithmetic “(38 – 12) is 14” → “is 26” (plateau 38, PEEP 12; conclusion “elevated / low compliance” unchanged). All original animations, graphics, and slide order preserved.

Human review: Case 2 numbers are internally inconsistent (Peak–Plateau and driving-pressure values do not reconcile with the stated peak/plateau/PEEP) — left unchanged pending author confirmation.

Companion files: Teaching guide (Vent-mechanics-4.11.docx) and Learner handout (Ventilators-Hand-out.pdf) — reviewed, no changes. Added context source-cited: Amato NEJM 2015; ARDSNet NEJM 2000; ATS/ESICM/SCCM 2017; ESICM ARDS 2023; Global ARDS definition 2024. Medical content should be clinician-reviewed before teaching.